HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F79MA
Statistical Models A

Semester 1 - 2020/21

Duration: 2 hours

Question	Marks
1	14
2	16
3	13
4	17
Total Marks	60

Candidates must attempt ALL FOUR questions.

ALL answers will be used for assessment.

To receive full credit you must show your work and explain your answers.

A University approved calculator may be used.

Statistical tables are provided at the end of the exam.

1. Let $\underline{\mathsf{X}} = (X_1, \ldots, X_n)$ be an i.i.d. random sample from a Weibull distribution, with probability density function given by

$$f(x;\lambda) = \frac{k}{\lambda}\left(\frac{x}{\lambda}\right)^{k-1} \exp\{-(x/\lambda)^k\}, \quad x \in \mathbb{R},$$

where $k > 0$ is a known shape parameter, and λ is an unknown scale parameter taking values in $\mathbb{R}^+$.

Consider the parametrisation $\theta = \lambda^k$.

a) Derive the likelihood function $L(\theta; \underline{\mathsf{X}})$.

[2 Marks]

b) Use the factorisation theorem to show that $\sum_{i=1}^n X_i^k$ is a sufficient statistic for θ.

[2 Marks]

c) Derive the maximum likelihood estimator $\hat{\theta}(\underline{\mathsf{X}})$ for θ.

[4 Marks]

d) Show that $\hat{\theta}(\underline{\mathsf{X}})$ is an unbiased estimator for θ.

[3 Marks]

e) Show that $\mathrm{Var}\{\hat{\theta}(\underline{\mathsf{X}})\} = \theta^2/n$.

[3 Marks]

[Total 14 Marks]

[PLEASE TURN OVER]

2. Let $\underline{\mathrm{X}} = (X_1, \ldots, X_n)$ be an i.i.d. random sample from an exponential distribution, with probability density function given by

$$f(x;\theta) = \lambda \exp\{-\lambda x\}, \quad x > 0\,,$$

where λ is an unknown parameter taking values in $\mathbb{R}^+$.

a) Derive the likelihood function $L(\lambda; \underline{\mathrm{X}})$. [2 Marks]

b) Using the result of part (a) above, derive the Fisher information $I(\lambda)$ measuring the amount of information that $\underline{\mathrm{X}}$ carries about λ.

[2 Marks]

c) Show that the Jeffreys' prior for λ is $\pi_J(\lambda) \propto \lambda^{-1}$).

[1 Mark]

d) Using the prior specified in part (c) above, derive the posterior distribution for λ. Report the mean and the variance of $\lambda|\underline{\mathrm{X}}$.

[4 Marks]

e) Now consider the Bayesian estimator $\hat{\lambda}(\underline{\mathrm{X}}) = \mathrm{E}(\lambda|\underline{\mathrm{X}})$ from a classical estimation viewpoint and show that $\hat{\lambda}(\underline{\mathrm{X}})$ is a biased estimator, with bias $B(\lambda) = \lambda/(n-1)$.

[5 Marks]

f) Using the Cramer Rao Lower Bound for biased estimators, calculate a lower bound for the variance of the estimator $\hat{\lambda} = \mathrm{E}(\lambda|\underline{\mathrm{X}})$.

[2 Marks]

[Total 16 Marks]

3. Suppose that following privatisation, an insurance company is going to obtain a listing on Stock Exchange. According to regulation, they need to provide evidence that the average annual return μ exceeds 150. Suppose the company has been established for ten years. To provide such evidence, they randomly select the annual return of 5 years and calculate a sample mean of $\bar{x} = 160$. Assume that the annual return has a normal distribution with known standard deviation $\sigma = 100$. (All units are in $£1000$).

Consider the following hypotheses

$$H_0 : \mu = 150 \qquad \text{v.s.} \qquad H_1 : \mu > 150\,,$$

and the test with rejection region given by

$$R = \left\{ \underline{x} : \sum_{i=1}^{n} x_i > k \right\}$$

where k is set to achieve a desired size α.

a) What does it mean to say that the considered likelihood ratio test is the uniformly most powerful test? [2 Marks]

b) Show that the family of considered likelihoods $\{L(\mu; \underline{X}) : \mu \in \mathbb{R}\}$ satisfy the Monotonic Likelihood Ratio Property (MLRP). [3 Marks]

c) Calculate the value of k such that the test has size $\alpha = 0.05$. [3 Marks]

d) Using the standard normal distribution table in page 6, or otherwise, calculate the power of the test derived in (c) for $\mu = 300$. [2 Marks]

e) Explain what is meant by the p-value associated with the test. Find the p-value for your randomly selected claims, and briefly discuss your result. [3 Marks]

[Total 13 Marks]

[PLEASE TURN OVER]

4. An insurance company is going to launch a new line of business and has to estimate the claim frequency for the associated policies (*i.e.*, the expected number of claims arising in the coming year). Based on past experience, it is believed that the number of claims each year follows a Poisson distribution with mean λ. The company does not know the value of λ, but based on general practice in the industry, it assumes that $\lambda \sim \text{Gamma}(\alpha, \beta)$ for some fixed values $\alpha, \beta > 0$.

a) What is the Bayes estimate of λ with respect to a quadratic loss function? Find this estimate.

[5 marks]

b) Write the posterior mean in a credibility form and determine the credibility factor.

[3 marks]

c) What is a credibility factor? Interpret your result in the following two settings: (i) when there is no data available, i.e. $n = 0$; and (ii) when the only information that you have is data from the risk.

[**Hint**: you should explain how the factor changes w.r.t. n and (α, β).]

[4 marks]

d) The data below show the aggregate claims in five successive years from 4 flood insurance policies in the portfolio of an insurer. These amounts have been adjusted to remove the effects of inflation and are given in units of $\pounds 1,000$.

	Year (j)						
Policy (i)	1	2	3	4	5	$\bar{X}_i$	$\sum_{j=1}^{n}(X_{ij} - \bar{X}_i)^2/(n-1)$
1	90	80	78	95	100	88.6	89.8
2	60	70	85	90	50	71	280
3	45	73	30	94	84	65.2	722.7
4	35	84	90	60	70	67.8	474.2

Calculate the credibility premium for the coming years for policy number 1 using EBCT Model 1.

[5 marks]

[Total 17 Marks]

Appendix:

x	$\Phi(x)$	x	$\Phi(x)$	x	$\Phi(x)$	x	$\Phi(x)$	x	$\Phi(x)$
0.00	0.50000	**0.40**	0.65542	**0.80**	0.78814	**1.20**	0.88493	**1.60**	0.94520
0.01	0.50399	**0.41**	0.65910	**0.81**	0.79103	**1.21**	0.88686	**1.61**	0.94630
0.02	0.50798	**0.42**	0.66276	**0.82**	0.79389	**1.22**	0.88877	**1.62**	0.94738
0.03	0.51197	**0.43**	0.66640	**0.83**	0.79673	**1.23**	0.89065	**1.63**	0.94845
0.04	0.51595	**0.44**	0.67003	**0.84**	0.79955	**1.24**	0.89251	**1.64**	0.94950
0.05	0.51994	**0.45**	0.67364	**0.85**	0.80234	**1.25**	0.89435	**1.65**	0.95053
0.06	0.52392	**0.46**	0.67724	**0.86**	0.80511	**1.26**	0.89617	**1.66**	0.95154
0.07	0.52790	**0.47**	0.68082	**0.87**	0.80785	**1.27**	0.89796	**1.67**	0.95254
0.08	0.53188	**0.48**	0.68439	**0.88**	0.81057	**1.28**	0.89973	**1.68**	0.95352
0.09	0.53586	**0.49**	0.68793	**0.89**	0.81327	**1.29**	0.90147	**1.69**	0.95449
0.10	0.53983	**0.50**	0.69146	**0.90**	0.81594	**1.30**	0.90320	**1.70**	0.95543
0.11	0.54380	**0.51**	0.69497	**0.91**	0.81859	**1.31**	0.90490	**1.71**	0.95637
0.12	0.54776	**0.52**	0.69847	**0.92**	0.82121	**1.32**	0.90658	**1.72**	0.95728
0.13	0.55172	**0.53**	0.70194	**0.93**	0.82381	**1.33**	0.90824	**1.73**	0.95818
0.14	0.55567	**0.54**	0.70540	**0.94**	0.82639	**1.34**	0.90988	**1.74**	0.95907
0.15	0.55962	**0.55**	0.70884	**0.95**	0.82894	**1.35**	0.91149	**1.75**	0.95994
0.16	0.56356	**0.56**	0.71226	**0.96**	0.83147	**1.36**	0.91309	**1.76**	0.96080
0.17	0.56749	**0.57**	0.71566	**0.97**	0.83398	**1.37**	0.91466	**1.77**	0.96164
0.18	0.57142	**0.58**	0.71904	**0.98**	0.83646	**1.38**	0.91621	**1.78**	0.96246
0.19	0.57535	**0.59**	0.72240	**0.99**	0.83891	**1.39**	0.91774	**1.79**	0.96327
0.20	0.57926	**0.60**	0.72575	**1.00**	0.84134	**1.40**	0.91924	**1.80**	0.96407
0.21	0.58317	**0.61**	0.72907	**1.01**	0.84375	**1.41**	0.92073	**1.81**	0.96485
0.22	0.58706	**0.62**	0.73237	**1.02**	0.84614	**1.42**	0.92220	**1.82**	0.96562
0.23	0.59095	**0.63**	0.73565	**1.03**	0.84849	**1.43**	0.92364	**1.83**	0.96638
0.24	0.59483	**0.64**	0.73891	**1.04**	0.85083	**1.44**	0.92507	**1.84**	0.96712
0.25	0.59871	**0.65**	0.74215	**1.05**	0.85314	**1.45**	0.92647	**1.85**	0.96784
0.26	0.60257	**0.66**	0.74537	**1.06**	0.85543	**1.46**	0.92785	**1.86**	0.96856
0.27	0.60642	**0.67**	0.74857	**1.07**	0.85769	**1.47**	0.92922	**1.87**	0.96926
0.28	0.61026	**0.68**	0.75175	**1.08**	0.85993	**1.48**	0.93056	**1.88**	0.96995
0.29	0.61409	**0.69**	0.75490	**1.09**	0.86214	**1.49**	0.93189	**1.89**	0.97062

Figure 1: Standard normal distribution table.

END OF PAPER